GATE ME 90 Days Challenge

Day 43/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Heat Transfer	**Fins & Unsteady Conduction**	**61 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on the fin equation, long and insulated-tip fins, corrected length, fin efficiency and effectiveness, lumped-capacitance (Biot number, time constant), Fourier number, Heisler charts and the semi-infinite solid. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The basic assumption of the lumped capacitance method of transient conduction is that

(A) the surface temperature is constant

(B) the heat transfer coefficient is zero

(C) the temperature of the body is uniform in space at every instant and changes only with time

(D) the body is semi-infinite

Q2 NAT

A long steel cylinder of 40 mm diameter (k = 15 W/m·K) is cooled in a fluid with h = 60 W/m²·K. The Biot number based on the lumped characteristic length (round off to two decimal places) is _____

Q3 NAT

A copper sphere of 30 mm diameter (ρ = 8900 kg/m³, c = 385 J/kg·K, k = 400 W/m·K) at 200 °C is cooled in air at 25 °C with h = 25 W/m²·K. The time for it to reach 80 °C (in s, round off to the nearest integer) is _____

Q4 MCQ

To make a thermocouple respond faster to temperature changes in a gas stream, one should

(A) use a larger bead

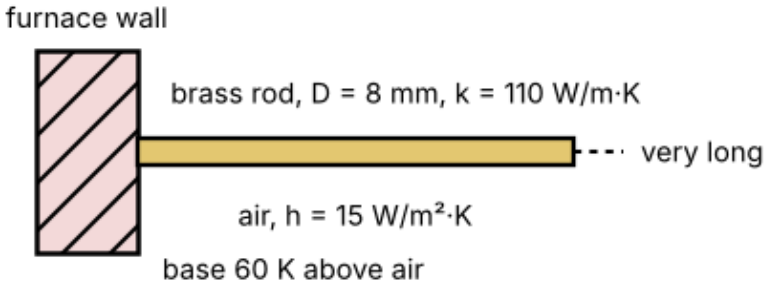
(B) use a smaller bead and increase the gas velocity

(C) use a bead material of higher density and specific heat

(D) reduce the gas velocity

Q5 NAT

A very long brass rod of 8 mm diameter (k = 110 W/m·K) projects from a furnace wall into air with h = 15 W/m²·K, as shown. Its base is 60 °C above the air temperature. The heat lost by the rod (in W, round off to two decimal places) is _____



Q6 MCQ

Fin effectiveness is defined as the ratio of

(A) heat transfer from the fin to the heat that would be transferred from the base area alone without the fin

(B) heat transfer from the fin to the heat it would transfer if the whole fin were at base temperature

(C) fin surface area to base area

(D) fin length to fin thickness

Q7 NAT

A straight rectangular aluminium fin (k = 200 W/m·K) is 2 mm thick and 20 mm long; h = 50 W/m²·K. Using the corrected length $L_c = L + t/2$ with an insulated-tip formula, the fin efficiency (in %, round off to one decimal place) is _____

Q8 MCQ

In a very long fin, the temperature excess $\theta = T - T_\infty$ varies along the fin length x as

(A) $\theta = \theta_b(1 - mx)$

(B) $\theta = \theta_b \cosh(mx)$

(C) $\theta = \theta_b/(1 + mx)$

(D) $\theta = \theta_b e^{-mx}$

Q9 NAT

A steel ball of 40 mm diameter (ρ = 7800 kg/m³, c = 460 J/kg·K, k = 40 W/m·K) at 420 °C is placed in air at 20 °C with h = 40 W/m²·K. The heat it loses in the first 300 s (in kJ, round off to two decimal places) is _____

Q10 MCQ

The one-term approximation used in the Heisler charts for transient conduction is accurate (within about 2 %) when

(A) Bi < 0.1

(B) the Fourier number Fo > 0.2

(C) the Fourier number Fo < 0.05

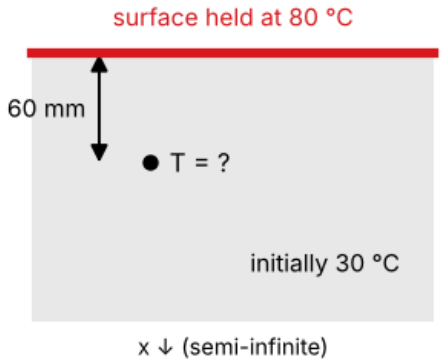
(D) Bi > 100 only

Q11 NAT

A large steel plate 50 mm thick ($\alpha = 1.2 \times 10^{-5}$ m²/s) is suddenly exposed to hot gases on both faces. The Fourier number based on the half-thickness after 10 minutes (round off to two decimal places) is _____

Q12 MCQ

A thick concrete slab ($\alpha = 5 \times 10^{-7}$ m²/s) is initially at 30 °C throughout. Its surface is suddenly raised to and held at 80 °C, as shown. Taking erf(0.5) = 0.5205, the temperature 60 mm below the surface after 2 hours is about



(A) 56.0 °C
(B) 26.0 °C
(C) 54.0 °C
(D) 46.0 °C

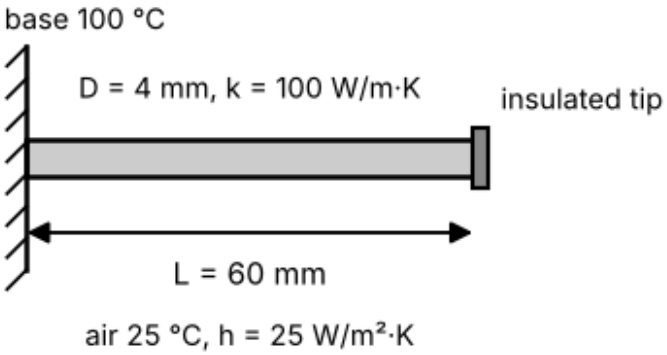
Q13 MCQ

A tube in a heat exchanger carries water inside, while air flows over its outside. To increase the heat transfer most effectively, fins should be attached

(A) on the water side (inside)
(B) on the air side (outside)
(C) equally on both sides
(D) on neither side, since fins always reduce heat transfer

Q14 NAT

An aluminium pin fin (k = 100 W/m·K) of 4 mm diameter and 60 mm length has an insulated tip, as shown. Its base is at 100 °C and the surrounding air is at 25 °C with h = 25 W/m²·K. The tip temperature (in °C, round off to one decimal place) is _____



Q15 MCQ

Two solid spheres of the same material, of diameters D and 2D, are cooled from the same initial temperature in the same fluid (lumped behaviour). The time taken by the larger sphere to reach a given temperature is

(A) the same as for the smaller sphere
(B) twice that of the smaller sphere
(C) four times that of the smaller sphere
(D) half that of the smaller sphere

Answer Key

Q	Type	Answer
Q1	MCQ	(C) the temperature of the body is uniform in space at every instant and changes only with time
Q2	NAT	0.04
Q3	NAT	793 (accept 787 to 799)
Q4	MCQ	(B) use a smaller bead and increase the gas velocity
Q5	NAT	2.74 (accept 2.71 to 2.77)
Q6	MCQ	(A) heat transfer from the fin to the heat that would be transferred from the base area alone without the fin
Q7	NAT	96.5 (accept 96.2 to 96.7)
Q8	MCQ	(D) $\theta = \theta_b e^{-mx}$
Q9	NAT	18.97 (accept 18.78 to 19.16)
Q10	MCQ	(B) the Fourier number Fo > 0.2
Q11	NAT	11.52 (accept 11.40 to 11.64)
Q12	MCQ	(C) 54.0 °C
Q13	MCQ	(B) on the air side (outside)
Q14	NAT	75.5 (accept 75.0 to 76.0)
Q15	MCQ	(B) twice that of the smaller sphere

Detailed Solutions

Q1 **Answer: (C) the temperature of the body is uniform in space at every instant and changes only with time**

Concept: When internal conduction resistance is negligible (Bi < 0.1), spatial temperature differences inside the body are small.

Then T = T(t) only, and an energy balance gives $\rho V c\, dT/dt = -hA_S(T - T_\infty)$.

Solution: $\theta/\theta_i = \exp(-t/\tau)$ with $\tau = \rho Vc/(hA_S)$.

Exam tip: Always check $Bi = hL_c/k$ with $L_c = V/A_S$ before using the lumped formula.

Q2 **Answer: 0.04**

Concept: Lumped Bi uses $L_c = V/A_S$; for a long cylinder $L_c = \pi R^2 L/(2\pi RL) = R/2$.

$L_c = 0.02/2 = 0.01$ m

$Bi = hL_c/k = 60 \times 0.01/15 =$ **0.04** (< 0.1, so lumped analysis is valid)

Exam tip: Heisler charts use $L_c = R$ (not R/2) for cylinders; do not mix the two definitions.

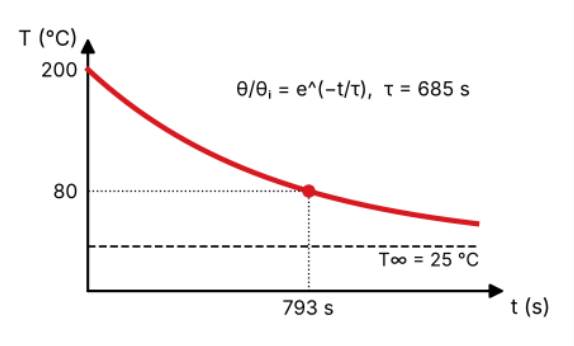
Q3 Answer: 793 (accept 787 to 799)

Concept: Bi = h(D/6)/k = 25 × 0.005/400 = 0.0003 < 0.1, so lumped: $t = \tau \ln(\theta_i/\theta)$.

$\tau = \rho c L_c/h = 8900 \times 385 \times 0.005/25 = 685.3$ s

$t = 685.3 \times \ln(175/55) = 685.3 \times 1.1575 =$ **793 s**

Exam tip: Work with temperature excesses $\theta = T - T_\infty$, never with T directly.



Q4 Answer: (B) use a smaller bead and increase the gas velocity

Concept: Time constant $\tau = \rho cV/(hA_s) = \rho cD/(6h)$ for a spherical bead.

A smaller D reduces τ directly; a higher gas velocity raises h and reduces τ further.

Higher ρc (option C) or lower velocity (option D) would slow the response.

Exam tip: After one τ the bead has covered 63.2 % of the step change; after 5τ, over 99 %.

Q5 Answer: 2.74 (accept 2.71 to 2.77)

Concept: Infinitely long fin: $Q = \sqrt{(hPkA_c)}\,\theta_b$, with the temperature excess falling as $\theta = \theta_b e^{-mx}$.

$P = \pi D = 0.02513$ m; $A_c = \pi D^2/4 = 5.027 \times 10^{-5}$ m²

$\sqrt{(15 \times 0.02513 \times 110 \times 5.027 \times 10^{-5})} = \sqrt{(2.084 \times 10^{-3})} = 0.04566$ W/K

$Q = 0.04566 \times 60 =$ **2.74 W**

Exam tip: $m = \sqrt{(4h/(kD))} = 8.26$ m⁻¹ here, so a rod longer than about 3/m ≈ 0.36 m already acts as 'infinite'.

Q6 Answer: (A) heat transfer from the fin to the heat that would be transferred from the base area alone without the fin

Concept: Effectiveness $\varepsilon_f = Q_{fin}/(hA_c\theta_b)$, the gain over the bare base.

Option (B) is the definition of fin efficiency η_f, a different quantity.

Fins are justified only when ε_f is well above 1 (≥ 2 in practice).

Exam tip: For a long fin, $\varepsilon_f = \sqrt{(kP/(hA_c))}$, so thin, closely spaced, high-k fins in gases work best.

Q7 Answer: 96.5 (accept 96.2 to 96.7)

Concept: For a thin rectangular fin (width ≫ t), $m = \sqrt{(2h/(kt))}$; the convecting tip is handled by extending the length by t/2.

$m = \sqrt{(2 \times 50/(200 \times 0.002))} = \sqrt{250} = 15.81$ m⁻¹; $L_c = 20 + 1 = 21$ mm

$mL_c = 0.332$; $\eta = \tanh(0.332)/0.332 =$ **96.5 %**

Exam tip: For a pin fin the corrected length is L + D/4.

Q8 **Answer: (D) $\theta = \theta_b e^{-mx}$**

Concept: The fin equation $d^2\theta/dx^2 - m^2\theta = 0$ has solution $\theta = C_1 e^{mx} + C_2 e^{-mx}$.
For an infinitely long fin $\theta \to 0$ as $x \to \infty$, so $C_1 = 0$ and $C_2 = \theta_b$.

Hence $\theta = \boldsymbol{\theta_b e^{-mx}}$, an exponential decay.

Exam tip: For an insulated tip, $\theta/\theta_b = \cosh m(L - x)/\cosh mL$.

Q9 **Answer: 18.97 (accept 18.78 to 19.16)**

Concept: $Bi = h(D/6)/k = 40 \times 0.00667/40 = 0.0067$, so lumped. Heat lost in time t: $Q = mc\theta_i(1 - e^{-t/\tau})$.

$m = 7800 \times \pi(0.04)^3/6 = 0.2614$ kg; $\tau = \rho c L_c/h = 7800 \times 460 \times 0.00667/40 = 598$ s

$mc\theta_i = 0.2614 \times 460 \times 400 = 48.10$ kJ; $1 - e^{-300/598} = 0.3945$

$Q = 48.10 \times 0.3945 =$ **18.97 kJ**

Exam tip: $mc\theta_i$ is the maximum heat the body can ever lose (as $t \to \infty$).

Q10 **Answer: (B) the Fourier number Fo > 0.2**

Concept: The exact solution is an infinite series in $\exp(-\lambda_n^2 Fo)$; higher terms decay quickly as Fo grows.
For Fo > 0.2 only the first term matters, and the charts (θ_0/θ_i versus Fo with 1/Bi as parameter) are built on it.
Bi < 0.1 is the lumped criterion, a different simplification.

Exam tip: For very short times (small Fo) treat the body as a semi-infinite solid instead.

Q11 **Answer: 11.52 (accept 11.40 to 11.64)**

Concept: $Fo = \alpha t/L^2$, with L the half-thickness for a plate heated on both sides.
L = 0.025 m, t = 600 s
$Fo = 1.2 \times 10^{-5} \times 600/0.025^2 =$ **11.52**

Exam tip: Fo > 0.2 here, so the one-term (Heisler) solution applies.

Q12 **Answer: (C) 54.0 °C**

Concept: Semi-infinite solid with a step change in surface temperature: $(T - T_s)/(T_i - T_s) = \text{erf}[x/(2\sqrt{(\alpha t)})]$.
$\sqrt{(\alpha t)} = \sqrt{(5 \times 10^{-7} \times 7200)} = 0.06$ m, so $x/(2\sqrt{(\alpha t)}) = 0.06/0.12 = 0.5$
$T = 80 + (30 - 80) \times 0.5205 = 80 - 26.0 =$ **54.0 °C**

Exam tip: At the surface erf(0) = 0 gives $T = T_s$; deep inside erf → 1 gives $T = T_i$.

Q13 **Answer: (B) on the air side (outside)**

Concept: The overall rate is limited by the larger resistance 1/(hA). Gas-side h (10–100 W/m²·K) is far lower than water-side h (thousands).
Adding area where h is low cuts the controlling resistance.
This is why car radiators and AC condensers have fins on the air side.

Exam tip: Fins pay off where hA is smallest.

Q14 **Answer: 75.5 (accept 75.0 to 76.0)**

Concept: Insulated tip: $\theta(x)/\theta_b = \cosh m(L - x)/\cosh mL$, so at the tip $\theta_L = \theta_b/\cosh mL$.

$m = \sqrt{(4h/(kD))} = \sqrt{(100/0.4)} = 15.81\ m^{-1}$; $mL = 0.949$; $\cosh 0.949 = 1.485$

$\theta_L = 75/1.485 = 50.5$ K, so $T_{tip} = 25 + 50.5 =$ **75.5 °C**

Exam tip: A tip far hotter than the air means the fin is short (mL small) and highly efficient.

Q15 **Answer: (B) twice that of the smaller sphere**

Concept: For the same θ/θ_i, $t \propto \tau = \rho c L_c/h$ with $L_c = D/6$.

Doubling D doubles L_c and hence τ.

So the larger sphere takes **twice** as long.

Exam tip: In the non-lumped (Heisler) regime, time at equal Fo scales with D^2 instead.

What next?

Day 43 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 44/90: Heat Transfer, Convection. Comment **GATE90** on our reel to get all links.
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